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National University of Computer and Emerging Sciences, Lahore Campus

Course:	Feedback Control System	Course Code:	EE 304
Program:	BS (Electrical Engineering)	Semester:	Fall 2016
Duration:	180 Minutes	Total Marks:	100
Paper Date:	14-Dec-16	Weight	50%
Section:	ALL	Page(s):	3
Exam:	Final Exam (Dr. Anjum Ali)	Roll No:	

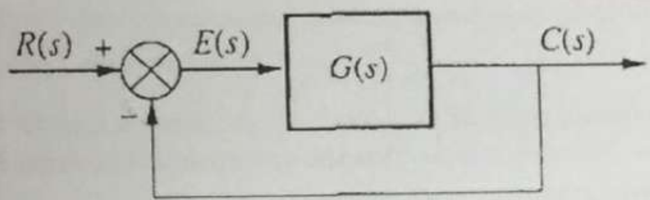
Instruction/Notes: **THIS EXAM HAS 10 QUESTIONS. Read ALL questions first. BOX ALL YOUR ANSWERS. Work at a speed of 15 min per question.**

1. **(10 points)** A control system with a closed loop transfer function has a characteristic polynomial given by

$$T(s) = s^5 + 7s^4 + 6s^3 + 42s^2 + 8^s + 56$$

Apply the Routh-Hurwitz criterion and find out how many poles of the system are in the right half plane. Is the system stable? Why?

2. **(10 points)** Consider the unity feedback system shown below:



If $G(s) = K.\frac{s^2+2s+7.5}{s(s+1)(s+2)(s+10)}$ and a root locus is plotted for the closed loop system for $0 \le K \le \infty$, answer the following questions:

(a) What is the real axis intercept of the asymptotes, and the angles of the asymptotes with the real axis?

(b) What are the angles of arrival for complex zeros?

3. **(10 points)** For the system of problem 2, if $G(s) = \frac{K}{(s+2)(s+3)(s+20)}$ and a root locus is plotted for the closed loop system for $0 \le K \le \infty$, answer the following questions:

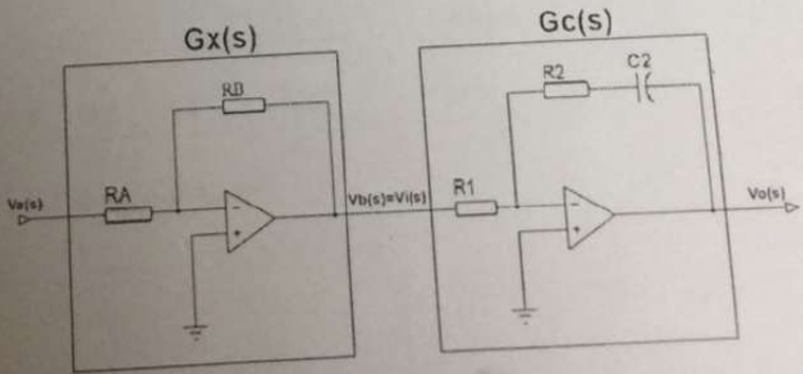
(a) Find the break-away point and the break-in point using differentiation.

(b) Calculate the jw axis crossings of the root locus.

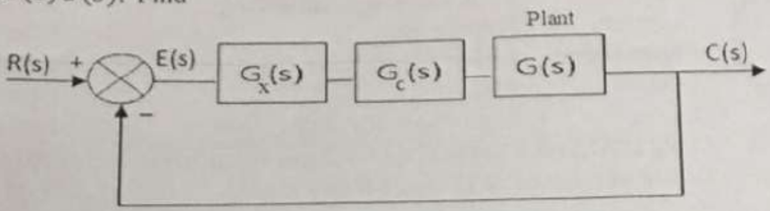
4. **(10 points)** The circuit shown below in the $G_x(s)$ and the $G_c(s)$ boxes implements a proportional compensator in cascade with a P.I. compensator.

(a) If $R_B = 10K$ ohms, find the value of R_A to get $G_x(s) = -(100)$.

(b) Use $R_1 = 20K$ and find the values of R_2 and C_2 so that $Gc(s) = -\frac{1}{100}\cdot\frac{s+0.15}{s}$



5. **(10 points)** The compensator circuit of question 4 is connected in cascade with a plant having $G(s) = \frac{1}{(s+1)(s+2)}$, as shown below. The forward path transfer function is given by $F(s) = Gx(s)\ Gc(s)G(s)$. Find

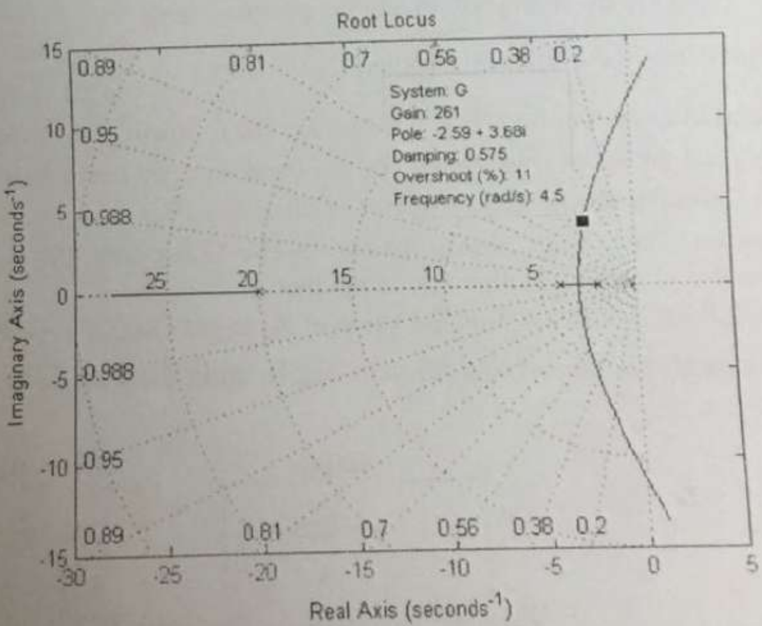


(a) the three static error constants Kp, Kv, and Ka for the compensated system.
(b) the steady state error for a unit step input, a unit ramp input and a unit parabola input after compensation.

6. **(10 points)** Given $G(s) = \frac{K}{(s^2+2s+2)}$, find the magnitude M of G(jw) in decibels, and phase P of G(jw) in degrees when $K = 1$ and $w = 3$ radians/sec.

7. **(10 points)** The root locus plot of an uncompensated negative unity feedback control system with an open loop transfer function $G(s) = \frac{K}{(s+2)(s+4)(s+20)}$ is shown below. When $K = 261$, the damping ratio ζ=0.575, and the dominant poles are at -2.59±j3.68, as shown in the root locus plot. It is required to design a **LAG** compensator for the system, **so that the new steady state error for a unit step input is half that *of the uncompensated system*.**

(a) Find the location of the compensator zero, z_c, assuming that the location of the compensator pole is at p_c = -0.001.
(b) Also, give the value of the steady state error for the compensated and uncompensated systems. Is the second order approximation valid for the compensated system? Why or why not?



8. **(10 points)** For the UNCOMPENSATED system of question # 7, compute the following:
   (a) Settling time, T_s.
   (b) Peak time, T_p.
   (c) Natural frequency, ω_n.
   (d) Damped frequency, ω_d.
   (e) % Overshoot.

9. **(10 points)** The state-space representation of a control system is given by:
$\frac{d\mathbf{x}}{dt} = \mathbf{Ax} + \mathbf{Bu}$ **and** $\mathbf{y} = \mathbf{Cx} + \mathbf{Du}$. It is known that $\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$, $\mathbf{C} = [2\ 7\ 1]$ and $\mathbf{D} = 0$.
Write the transfer function for this system as a ratio of two polynomials.

10. **(10 points)** Write the differential equation that is mathematically equivalent to the block diagram shown below. Assume $r(t) = 3t^3$.

$$R(s) \longrightarrow \boxed{\frac{s^5 + 2s^4 + 4s^3 + s^2 + 4}{s^6 + 7s^5 + 3s^4 + 2s^3 + s^2 + 5}} \longrightarrow C(s)$$

==================== END OF EXAM ========================

REMEMBER TO BOX ALL YOUR FINAL ANSWERS.

Some hints:

TABLE 9.7 Types of cascade compensators

Function	Compensator	Transfer functi
Improve steady-state error	PI	$K\frac{s+z_c}{s}$
Improve steady-state error	Lag	$K\frac{s+z_c}{s+p_c}$
Improve transient response	PD	$K(s+z_c)$
Improve transient response	Lead	$K\frac{s+z_c}{s+p_c}$
Improve steady-state error and transient response	PID	$K\frac{(s+z_{lag})(s+}{s}$

$$\boxed{\zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2 + \ln^2(\%OS/100)}}}$$

$$\boxed{T_s = \frac{4}{\zeta\omega_n}} \qquad \boxed{T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}}$$

$$\boxed{\%OS = e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100}$$

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1} \qquad (4.24)$$